

1 Assignment: Wiener Filtering

Consider the autoregressive process described by the difference equation

$$d(n) = 0.75d(n-1) + v(n) \quad (1)$$

and the noisy process

$$u(n) = d(n) + w(n) + 0.5w(n-1), \quad (2)$$

where both $v(n)$ and $w(n)$ are white noise with zero mean and unit standard deviation, and $\mathbb{E}[v(n)w(n)] = 0$. We are interested in a **first-order Wiener filter** estimating $d(n)$ from $u(n)$.

1.1 Draw the block diagram of the system

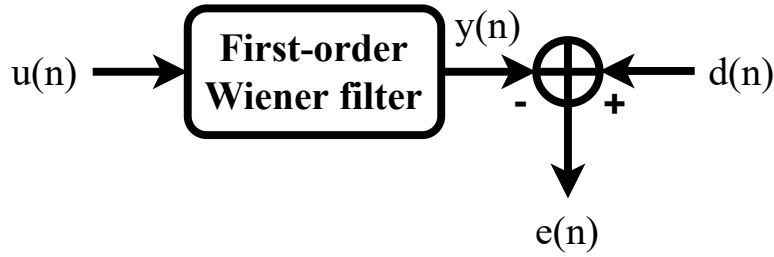


Figure 1: Block diagram of the system.

In the above diagram, $y(n) = \hat{d}(n)$.

1.2 Write the error signal $e(n)$ and the cost function to be minimized

The error signal is

$$\begin{aligned} e(n) &= d(n) - \hat{d}(n) \\ &= d(n) - \sum_{i=0}^1 \omega_i u(n-i) \\ &= d(n) - \omega_0 u(n) - \omega_1 u(n-1). \end{aligned} \quad (3)$$

The cost function to be minimized is

$$\begin{aligned} \mathcal{J} &= \mathbb{E}[e^2(n)] \\ &= \mathbb{E}[(d(n) - \omega_0 u(n) - \omega_1 u(n-1))^2]. \end{aligned} \quad (4)$$

1.3 Calculate, by hand, expressions for the optimal set of weights as well as obtain the values of the optimal weights

To obtain expressions for the optimal weights, we just have to calculate the following partial derivatives:

$$\begin{aligned}\frac{\partial \mathcal{J}}{\partial \omega_0} &= -2\mathbb{E}[(d(n) - \omega_0 u(n) - \omega_1 u(n-1)) u(n)] = 0; \\ \frac{\partial \mathcal{J}}{\partial \omega_1} &= -2\mathbb{E}[(d(n) - \omega_0 u(n) - \omega_1 u(n-1)) u(n-1)] = 0.\end{aligned}\tag{5}$$

We can rewrite (5) as

$$\begin{aligned}\omega_0 r_u(0) + \omega_1 r_u(1) &= r_{du}(0); \\ \omega_0 r_u(1) + \omega_1 r_u(0) &= r_{du}(1).\end{aligned}\tag{6}$$

And, in matrix form:

$$\begin{bmatrix} r_u(0) & r_u(1) \\ r_u(1) & r_u(0) \end{bmatrix} \begin{bmatrix} \omega_0 \\ \omega_1 \end{bmatrix} = \begin{bmatrix} r_{du}(0) \\ r_{du}(1) \end{bmatrix},\tag{7}$$

which constitutes the Wiener-Hopf equations.

We then obtain values for the different correlation coefficients as follows:

$$\begin{aligned}r_u(0) &= \mathbb{E}[(d(n) + w(n) + 0.5w(n-1)) (d(n) + w(n) + 0.5w(n-1))] \\ &= r_d(0) + 1.25; \\ r_u(1) &= \mathbb{E}[(d(n) + w(n) + 0.5w(n-1)) (d(n-1) + w(n-1) + 0.5w(n-2))] \\ &= r_d(1) + 0.5; \\ r_{du}(0) &= \mathbb{E}[(0.75d(n-1) + v(n)) (d(n) + w(n) + 0.5w(n-1))] \\ &= 0.75r_d(1) + 1; \\ r_{du}(1) &= \mathbb{E}[(0.75d(n-1) + v(n)) (d(n-1) + w(n-1) + 0.5w(n-2))] \\ &= 0.75r_d(0).\end{aligned}\tag{8}$$

To get the final values, we also need to calculate numbers for the autocorrelation coefficients $r_d(0)$ and $r_d(1)$ as follows:

$$\begin{aligned}r_d(0) &= \mathbb{E}[(0.75d(n-1) + v(n)) (0.75d(n-1) + v(n))] \\ &= 0.5625r_d(0) + 1; \\ r_d(1) &= \mathbb{E}[(0.75d(n-1) + v(n)) (0.75d(n-2) + v(n-1))] \\ &= 0.5625r_d(1) + 0.75.\end{aligned}\tag{9}$$

From (9), it is straightforward to find that $r_d(0) = 2.2857$ and $r_d(1) = 1.7143$. Knowing this, we come back to (8): $r_u(0) = 3.5357$, $r_u(1) = 2.2143$, $r_{du}(0) = 2.2857$ and $r_{du}(1) = 1.7143$.

Finally, we handle (7) to achieve the values of the optimal weights as follows:

$$\begin{bmatrix} \omega_0 \\ \omega_1 \end{bmatrix} = \begin{bmatrix} r_u(0) & r_u(1) \\ r_u(1) & r_u(0) \end{bmatrix}^{-1} \begin{bmatrix} r_{du}(0) \\ r_{du}(1) \end{bmatrix} = \begin{bmatrix} 0.5640 \\ 0.1316 \end{bmatrix}. \quad (10)$$

1.4 Obtain a numerical approximation of the optimal weights. For that, using, e.g., MatLab or Python, generate 10,000 samples of $d(n)$ and $u(n)$, where $d(0)=w(0)=0$, and apply the expressions calculated in 1.3. Compare the result with that from 1.3

One run of my MatLab implementation provides the values $\hat{\omega}_0 = 0.5653$ and $\hat{\omega}_1 = 0.1342$. Notice that due to the stochastic nature of this estimation procedure, different runs will provide different weight values, which are approximations (in a least-squares sense) to the actual Wiener filter weights.

1.5 Using the estimated Wiener filter, apply it to $u(n)$ to calculate the estimate $\hat{d}(n)$. Both by hand and using numerical approximation, calculate the error variances $\sigma_{du}^2 = \mathbb{E}[(d(n) - u(n))^2]$ and $\sigma_{d\hat{d}}^2 = \mathbb{E}[(d(n) - \hat{d}(n))^2]$ and compare them

Similarly to what we did for 1.3,

$$\begin{aligned} \sigma_{du}^2 &= \mathbb{E}[(d(n) - u(n))^2] \\ &= \mathbb{E}[(-w(n) - 0.5w(n-1))(-w(n) - 0.5w(n-1))] \\ &= 1.25; \\ \sigma_{d\hat{d}}^2 &= \mathbb{E}[(d(n) - \hat{d}(n))^2] \\ &= \mathbb{E}[(d(n) - \omega_0 u(n) - \omega_1 u(n-1))(d(n) - \omega_0 u(n) - \omega_1 u(n-1))] \\ &= r_d(0) + 2(\omega_0 \omega_1 r_u(1) - \omega_0 r_{du}(0) - \omega_1 r_{du}(1)) + (\omega_0^2 + \omega_1^2) r_u(0) \\ &= 0.7709. \end{aligned} \quad (11)$$

One run of my MatLab implementation provides the following values for such error variances: $\hat{\sigma}_{du}^2 = 1.2423$ and $\hat{\sigma}_{d\hat{d}}^2 = 0.7692$. As above, due

to the stochastic nature of this estimation procedure, different runs will provide different error variance values, which are approximations, again in a least-squares sense, to the actual error variance values.

